Canadian Analyst Job Market

Salary & Demand Dashboard

DATA VISUALIZATION · CAPSTONE PROPOSAL

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01 Background & Motivation

The Canadian analyst job market has expanded with data-driven decision-making, yet graduates still lack a consolidated view of where roles are concentrated and what salaries to expect. Browsing Indeed.ca one posting at a time obscures the patterns that would inform a real career decision. This project consolidates 1,796 cleaned Canadian job postings across 14 provinces, 10 job-title categories, 22 industries, and 1,057 skill tags into one interactive dashboard. Salaries span $43,720 - $158,640 and vary by city, seniority, and work type.

02 Purpose & Objectives

Purpose: Turn fragmented job-board data into a decision tool that helps non-technical users identify the most realistic, best-paying analyst paths in Canada.

- ✓ Map geographic demand hotspots by province and city using a choropleth view.
- ✓ Identify the most in-demand skills across analyst roles and link them to salary impact.
- ✓ Apply course design principles: cognitive load, data-ink ratio, pre-attentive processing to every chart.

03 Key Questions

The dashboard is designed to answer five concrete questions:

Q1. *Which province offers the highest median salary for entry-level analyst roles?*

Q2. *How does seniority affect salary range spread within each industry?*

Q3. *Which cities concentrate the most analyst demand outside Ontario?*

04 Target Audience

Primary: recent graduates and career-changers in Ottawa deciding which roles to pursue.

Secondary: Lambton program advisors and recruiters benchmarking compensation. The dashboard assumes no SQL or BI background.

05 Preliminary Data Exploration

Ontario dominates postings (53%) but Alberta shows competitive salaries despite only 11% market share, invisible on a simple volume chart. Remote roles are scarce (8%), and the "ANY" seniority bucket covers 76% of postings.

Table 1. Field vs Key Statistics

Field	Key Statistics
Province	14 provinces · Ontario 53%, BC 14%, Alberta 11%
Job Title	10 categories · top: Senior Supply Chain (262), BI Analysts (229)
Seniority	ANY 76% · Senior 20% · Mid 2%
Work Type	In-Person 91% · Remote 8% · Hybrid 1%
Industry	22 types · Technology 19% · Healthcare 7% · Others 53%
Skills	1,057 unique tags · Excel, Power BI, Data Studio
Salary (Avg)	Mean $78,435 · range $43,720–$158,640 · std. dev. $18,027

06 Dashboard Layout & Visualizations

The dashboard follows an F-pattern layout: KPIs top-left, comparative charts middle, detail table at the bottom. Each chart is justified through the 3-Lens Framework (Purpose · Data · Audience).

6.1 Market Overview

KPI / BAN tiles: Median salary, total postings, top province, pre-attentive numerals, fastest scan.
Horizontal bar · job title × seniority: Length is the most accurate pre-attentive attribute for comparison.

6.2 Salary Analysis

Box plots · salary by job title: Show distribution and spread, protects against misleading means.
Scatter · min vs. max salary by industry: Reveals salary-band spread; caveat: correlation ≠ causation.

6.3 Skills & Interactivity

Tree map · top skills (size = frequency): Area reads more reliably than typographic size in a word cloud.
Global filter panel: Province, seniority, work type, industry, salary, grouped by proximity (Gestalt).

Sketch · Dashboard wireframe (low-fidelity)

<table>
<tr><th colspan="5">TITLE BAR · CANADIAN ANALYST JOB MARKET</th></tr>
<tr><td>KPI
Median</td><td>KPI
Postings</td><td>KPI
Top prov.</td><td colspan="2">CHOROPLETH MAP
Postings by province</td></tr>
<tr><td colspan="3">HORIZONTAL BAR
Job title × seniority</td><td colspan="2">BOX PLOT
Salary by job title</td></tr>
<tr><td colspan="2">FILTERS
province · seniority · work type · industry · salary</td><td colspan="2">TREEMAP
Top skills · frequency</td><td>DETAIL TABLE
Drill-down view</td></tr>
</table>

07 Research · Reference Example

We benchmarked against the Robert Half 2025 Canada Salary Guide, which publishes compensation ranges by city and seniority. It is a static PDF with no skill, work-type, or industry filtering. Our dashboard adds the multi-dimensional filtering and skill-frequency layer it cannot provide.

Source: https://www.roberthalf.com/ca/en/insights/salary-guide

08 Selected Tool

Power BI is the primary build platform; Google Looker Studio is the secondary target. Power BI handles the dataset, tree map, and multi-filter interactivity natively. Looker Studio is added because it publishes via URL with no account required.

09 Design Principles Applied

A single accent color marks the metric we want the audience to notice first; everything else stays neutral so data carries the message (data-ink ratio, per Tufte). Color is never the sole encoding, each comparison also varies position or label. Interactivity is limited to five filters (Hick's Law).

10 Scope & Constraints

IN SCOPE

- Interactive Power BI or Data Studio dashboard built on the cleaned dataset (no live collection).
- Five to seven visualizations across geography, salary, and skill frequency.

OUT OF SCOPE

› Real-time data collection, predictive modeling, and external API integration.

› Topic overlap with the BAM-2024 BI Tools project.

CONSTRAINTS

› The dataset is a historical snapshot and may not reflect the current market exactly.

› The "Others" industry category covers 53% of postings, limiting industry-level depth.